Supplementary Material: ConnectomeQuest: Certificate-Carrying Interaction under Costly Partial Observation

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Guide to the supplementary material

The supplement follows the claim order of the main paper: formal details, dataset and access provenance, policy and inference details, graph diagnostics, certificate-integrity tests, RGB validation, and reproduction.

C1 Formal details

For a fixed random tape, the closed-loop trajectory is

$$\mathcal{T} = (s_0, z_0, o_0, a_0, s_1, z_1, o_1, \ldots, a_{L-1}, s_L, z_L, o_L).$$

It is replay-valid when each observation equals the declared observation map, each state pair satisfies the controller and environment updates, and recorded costs and outcomes match replay. Replay validity is distinct from action admissibility: a charged API rejection can be replay-valid.

After anchor inspection, let $M(o)$ be the disclosed middle-node set. For the fixed controller, let $U_q(o) \subseteq M(o)$ contain the middle nodes from which an accepted certificate remains affordable. If $(m_1, \ldots, m_J)$ is the sequence of distinct branch openings, the first-acquired-certificate reciprocal rank is $1/j$ for the least j whose branch yields an accepted certificate, and zero if no branch does. Its sample mean is the first-acquired-certificate MRR. This execution metric is distinct from relevance of a candidate before execution.

C1.1 Dependency-indexed validation: proofs

Proof of Proposition 2. Fix a suffix position i whose support value changes. Interpolate from (A_t, θ_t) to (A_{t+1}, θ_{t+1}) by changing one coordinate from $W_t \cup N_t$ at a time. Because the endpoint support values differ, at least one single-coordinate step changes Supp_i. Receipt completeness places that coordinate in $D_i \cap W_t$ when it is a receipt. When it is a semantic key, the support predicate reads that key, and key completeness places it in $K_i \cap N_t$. Independently, every changed key read by the planner for step i is in $K_i \cap N_t$. The corresponding exact posting list therefore contains i, so $i \in \mathcal{J}(\Delta_t)$. Revalidating all positions in this set therefore includes every changed support decision. In particular, if the next checked action loses its final active support, its position is revalidated and the authorization rule rejects it. The reasoning uses issuer-relative receipt activity and makes no claim about unmodelled physical preconditions. $\square$

Proof of Proposition 3. Complete validation evaluates the entire post-update support vector. By Proposition 2, dependency-indexed validation evaluates every coordinate that can have changed; all others retain their exact cached values. Thus both obtain the same support vector. Exact inverse maps also make them agree whether $N_t \cap \bigcup_i K_i$ is empty. Applying the same retain-or-replan rule,

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Inputs. All methods receive the same remaining plan, complete receipt and semantic-key dependency records, active receipt identifiers, claim-to-receipt bindings, and one evidence update. The update contains withdrawn receipt identifiers, superseded receipt identifiers, changed semantic keys, and added claim bindings.

Full replanning. On every nonempty evidence update, discard the remaining plan and call the common deterministic planner.
Complete validation. On a nonempty update, visit every remaining plan position and evaluate its required claims. Call the common planner if any required claim lacks active receipt support or if the changed-key set intersects the union of the recorded planner-read semantic keys; otherwise retain the plan.
Dependency-indexed validation. Look up the posting lists of every withdrawn or superseded receipt identifier and every changed semantic key. Check only the retrieved remaining positions. Call the common planner if a retrieved required claim lacks active receipt support or a changed key has a retrieved planner-read posting; otherwise retain the plan.
Unchecked reuse. Continue the cached plan; it never checks or replans after an evidence update. This is a safety ablation, not a competent baseline.

Accounting. Record predicate evaluations, successor expansions, plan-position visits, index lookups, posting-list entries, planner calls, revalidation time, and planning time separately. Index construction is timed and included in validation-and-replanning CPU time.

Figure 1: The four access-matched revalidation procedures. Hashing and serialization are integrity instrumentation rather than decision steps.

they either retain the same suffix or invoke the same deterministic planner from the same state. Their next authorization decisions are therefore equal. □

Proof of Corollary 1. Constructing the two inverse maps enumerates each of the Z recorded dependencies once. For update t, expected-constant-time dictionary access contributes one lookup per member of $W_t \cup N_t$, and enumerating the retrieved posting lists contributes one operation per stored entry in the displayed sums. Summing construction and update terms gives the bound. Complete validation visits every remaining suffix position by definition. Neither count includes predicate evaluation, index maintenance, or a planner call; those quantities are measured separately. □

C1.2 Access-matched full-episode revalidation

C2 Data and access provenance

Table 1: Source assets and access metadata recorded in release manifests.

Asset	Version/configuration	Access source	License/terms
H01 C3	20210729-c3	Google Cloud public release	CC-BY-4.0
MANC	1.0 / male-cns:1.0	Janelia FlyEM files / neuPrint	CC-BY
HemiBrain	1.2.1 / hemibrain:1.2.1	neuPrint	CC-BY-4.0
MiniGrid/BabyAI	recorded package versions	Python dependencies	upstream licenses

Table 2: Versioned active-environment snapshots. Edge columns are disjoint: task semantic is the native query predicate; other semantic contains every remaining semantic edge; total is their sum with both wiring directions. Of 34,499 H01 neuron nodes, 33,752 have an official cortical-layer label; the remaining 747 are valid wiring nodes but cannot support the native semantic required claim.

Graph	neurons	nodes	wiring	inverse	task sem.	other sem.	total
H01 C3/20210729	34,499	34,506	232,211	232,211	33,752	0	498,174
MANC v1.0	22,744	26,637	3,874,922	3,874,922	22,744	0	7,772,588
HemiBrain v1.2.1	21,739	35,211	3,550,061	3,550,061	19,723	260,810	7,380,655

Only compact neuron-level connectivity and released annotations are retained. Raw EM imagery, meshes, skeletons, and per-synapse point tables are excluded. HemiBrain access uses a user-supplied credential that is never stored in an artifact. Source versions, URLs, filters, licenses, and file hashes are recorded in manifests. H01 is a small tissue sample and is not representative of human cortex generally; fly and human labels are not asserted to be homologous.

The primary task is typed two-hop witness search. CHECKEDGE may test a semantic edge only at the current focus after an acquired two-hop wiring path reaches that focus; it is therefore not an arbitrary membership oracle. For negated auxiliary queries, the environment may issue an explicit negative receipt against the immutable graph. Absence from a bounded page is never treated as evidence of non-existence.

C3 Policies and statistical procedure

The Attribute scorer has two hop-specific multilayer-perceptron heads of width 128, dropout 0.1, and 31 observable input coordinates. The coordinates comprise acquired count and weight summaries, relation and equality indicators, budget and frontier state, within-list order, and purchased-probe summaries. Numeric entity identifiers and entity embeddings are excluded. Training uses source graphs only, 20 epochs, AdamW, learning rate 3×10^{-4}, and domain–hop-balanced sampling. The exact coordinate transforms and normalizations are provided in the released configuration.

Table 3: Attribute-scorer feature groups. Numeric entity identifiers are not feature values; numeric tie-breaking can still affect rank-derived inputs and top-V. The paid-observation flag occurs in both network branches, so 31 coordinates represent 30 distinct scalars.

Group	Count	Observable quantities
Acquired evidence	6	log degree counts; incoming-weight sum and maximum; positive-edge fraction; log negative-receipt count
Equality and relation	8	current-node, memory, anchor, endpoint, and semantic-match indicators
Budget, frontier, order	7	budget and step fractions; frontier flag; local weight and fraction; reverse rank; top-four flag
Paid-observation flag	1	availability copied to the base network branch
Paid-observation summaries	9	the same availability gate plus percentile and robust-z transforms of four acquired statistics
Total	31	coordinates representing 30 distinct scalars

The Recurrent policy uses the same 31 candidate features, a 128-dimensional recurrent state, dropout 0.1, and a 32-dimensional query-family embedding that contains neither entity nor target identity. Its state resets at each ranking call. REINFORCE training uses source-graph decision records for 25 epochs and 16,000 trajectories per epoch, with entropy weight 0.01. Evaluation always uses the common controller and verifier.

Expert selector is a one-hidden-layer, 32-unit SiLU classifier that chooses among Weight, Random, Attribute, Recurrent, and a diversity ordering using only pre-action source-graph features. It is trained for 100 epochs with AdamW, batch size 128, learning rate 10^{-3}, and weight decay 10^{-4}; a held-out source split fixes its confidence threshold. Attribution in Table 7 prevents this optional component from being mistaken for the interaction interface.

The first anchor is the uncertainty cluster. Paired comparisons align query identifiers, average fixed-seed outcomes within query, and resample first-anchor clusters while retaining all queries in each sampled cluster. Intervals condition on fixed checkpoints, seeds, controller, and query set. The two planned fly-graph contrasts use a Bonferroni correction; all other audits are explicitly exploratory.

C4 Graph results and diagnostics

Table 4: Prospectively frozen evaluation on 1,000 previously unused query pairs per graph. Entries are certificate success/unconditional charged actions, averaged across seeds 17/29/43 for stochastic policies; Weight is deterministic. All policies use the same frozen controller and observations.

Graph	Weight	Random	Recurrent	Attribute	Expert selector	Random→Attribute
H01	0.4090/42.63	0.5100/39.82	0.4840/40.64	0.4117/41.27	0.3920/43.14	0.4017/42.05
MANC	0.0990/60.18	0.1050/60.11	0.1030/60.01	0.1107/59.96	0.1120/60.03	0.1003/60.19
HemiBrain	0.1270/59.52	0.1347/59.23	0.1437/58.86	0.1343/58.91	0.1483/58.83	0.1367/58.86

Table 5: First-anchor-cluster bootstrap for the held-out graph comparison. Intervals for the two planned fly-graph Expert-selector–Weight contrasts use a two-comparison Bonferroni correction; H01 is descriptive with a nominal 95% interval.

Graph	Contrast	Difference	Interval	Clusters
H01	Expert selector−Weight	-0.0170	$[-0.0295, -0.0048]$	837
MANC	Expert selector−Weight	+0.0130	$[-0.0094, +0.0360]$	807
HemiBrain	Expert selector−Weight	+0.0213	$[+0.0030, +0.0397]$	806

Table 6: Charged actions in the held-out graph evaluation, separated by outcome. The unconditional mean retains failed episodes; Exhaust. is the fraction reaching the controller's 63-action exploration limit (one transition is reserved for the zero-cost submission). These costs must be read jointly with success in Table 4.

Graph	Policy	n	All	Success	Failure	Exhaust.
H01	Weight	1000	42.63	31.07	50.63	0.352
H01	Random	3000	39.82	29.46	50.60	0.291
H01	Recurrent	3000	40.64	29.62	50.98	0.309
H01	Attribute	3000	41.27	28.18	50.42	0.345
H01	Expert selector	3000	43.14	31.20	50.83	0.365
H01	Random→Attribute	3000	42.05	29.31	50.61	0.348
MANC	Weight	1000	60.18	36.15	62.82	0.896
MANC	Random	3000	60.11	36.39	62.89	0.892
MANC	Recurrent	3000	60.01	35.22	62.86	0.893
MANC	Attribute	3000	59.96	36.05	62.93	0.887
MANC	Expert selector	3000	60.03	36.98	62.93	0.886
MANC	Random→Attribute	3000	60.19	35.60	62.93	0.897
HemiBrain	Weight	1000	59.52	36.50	62.87	0.868
HemiBrain	Random	3000	59.23	35.93	62.86	0.860
HemiBrain	Recurrent	3000	58.86	35.13	62.85	0.850
HemiBrain	Attribute	3000	58.91	33.45	62.86	0.860
HemiBrain	Expert selector	3000	58.83	35.73	62.86	0.846
HemiBrain	Random→Attribute	3000	58.86	33.57	62.87	0.858

The visible-witness ceiling uses privileged answer knowledge only to verify that a generated query contains an affordable witness in the disclosed pages. It is not an observable competitor.

The coarsening intervention replaces only the native semantic target by one of seven fixed, frequency-balanced target unions. Wiring, anchor, pairing identifier, policy checkpoint, and random seed remain fixed. These are controlled benchmark labels, not biological equivalence classes. Visible target prevalence is the mean query-wise fraction of relevant nodes among first-page two-hop candidates and is never exposed to a policy.

C5 Certificate-integrity audit

The verifier receives an untrusted certificate and a separately trusted query, graph/episode identity, current step, and issuance registry. It validates receipt bindings and graph-witness entailment without loading an answer set. Trusting a claimant-supplied registry would invalidate the guarantee.

The replay audit contains 60,000 graph-search episodes from four policies and three graphs. Every one of its 22,662 nonempty submissions is accepted; the remaining submissions are empty and hence contain no certificate to verify. A separate mutation diagnostic fixes 300 queries per graph and retains all 1,800 search episodes, including failures; only acquired certificates enter its mutation tests. Provenance/binding mutations remove receipts or alter episode, graph, query, or relation bindings. Structural mutations splice inconsistent middle nodes, answers, and semantic edges. These two cohorts are not pooled.

These deterministic rejection counts measure coverage of the stated mutation generator. Because mutations derived from the same acquired certificate are correlated and were not sampled from a probability model, we report no binomial confidence bound or attack-success estimate. Certificate soundness remains conditional on the trusted issuer and registry.

Table 7: Hop attribution on the held-out evaluation. Expert selector and Weight→Attribute share the Attribute second-hop scorer; only first-hop ordering differs.

Graph	Expert selector	Weight→Attribute	Success difference	Charged-action difference
H01	0.3920	0.3920	+0.0000	-0.00
MANC	0.1120	0.1120	+0.0000	+0.00
HemiBrain	0.1483	0.1483	+0.0000	+0.00

Table 8: Privileged visible-witness control on the same 5,000 native queries per graph. Oracle policies have full answer access; these are not observable competitors. Cost averages all episodes; maximum is for the visible-witness policy.

Graph	Historical	Visible witness	Cost historical	Cost visible	Max cost
H01	1.0000	1.0000	15.68	15.18	16
MANC	0.9118	1.0000	28.42	15.98	16
HemiBrain	0.9122	1.0000	28.99	15.99	16

C6 RGB retained-plan validation

The RGB study attempted 800 layouts, 200 in each of four requested remaining-plan-length strata. Eligibility was decided before any compared method ran: the layout generator had to succeed; the common RGB exploration controller had to reach a nonterminal staging state; the common planner had to return a complete mission plan in the requested stratum; the requested update schedule had to be constructible; and instrumentation had to finish without an exception. This method-independent rule retained 513 layouts. Four methods, three update seeds, and six schedule conditions produced 12,312 method–condition rows and 3,078 complete paired cells. Panel A contains 381 eligible layout–update–schedule cells from the long and very-long strata with one or two withdrawals and at least 80% irrelevant updates; Panel B contains one final-support-loss cell for each of the 513 retained layouts. These cohorts answer different questions and are not compared across panels. All methods share the planner, action model, pre-update trajectory, RGB-derived belief, and update schedule. Access hashes agree and every stored environment trace replays exactly.

Each public observation contains an egocentric 7×7 RGB image rendered at 8×8 pixels per tile, position relative to the episode start, exact cardinal direction, inventory object type, mission string, and the previous action acknowledgement, reward, and termination flag. Thus localization and heading are privileged public inputs; the study does not test visual odometry. The frozen 21,147-parameter tile classifier has two 3×3 convolutional layers (32 and 64 channels, SiLU activations), global average pooling, and a 27-class linear head. It was trained for 20 epochs on 4,548 renderer tiles with cross-entropy and AdamW (learning rate 3×10^{-3}, weight decay 10^{-4}, batch size 256, seed 17); accuracy on 1,707 held-out collected tiles was 0.9947. The checkpoint SHA-256 begins `092ea8aeef264421`. Training and evaluation use the same renderer, so we make no visual-domain-generalization claim. The classifier initializes semantic-map claims and their image-receipt bindings. The later receipt withdrawals are controlled protocol interventions shared by all methods, not learned detections of perception errors.

Primitive work reports predicate evaluations, planner-successor expansions, retained-plan positions, and index lookups separately; no arbitrary conversion factor is imposed. The prospectively frozen timing confirmation contains 221 eligible layouts and 1326 primary paired cells. Its primary population requires at least 64 remaining actions and four or eight distinct irrelevant receipt withdrawals. Dependency-indexed and complete validation agree on 100.0% of authorization decisions; the former reduces validation-and-replanning CPU time by 96.7% (layout-bootstrap 95% interval [96.4%, 96.9%]) and measured plan-position-plus-predicate work by 100.0%. This endpoint is the sum of index construction, index maintenance, required-claim validation, and planner calls; it excludes CNN inference, simulator stepping, serialization, and action execution. Across the 221 eligible layouts (median remaining-plan length 81), complete versus dependency-indexed median times were 7.036 versus 0.967 ms for one withdrawal, 28.220 versus 1.084 ms for four, and 60.611 versus 1.211 ms for eight. Measurements used one pinned logical CPU on an AMD EPYC 7763 host under Linux x86-64, with `OMP_NUM_THREADS=MKL_NUM_THREADS=OPENBLAS_NUM_THREADS=1`, cyclic method order, ten warmups, and forty retained repetitions per cell. The confirmation attempted 256 very-

Table 9: Paired coarsening on the same anchors and pairing identifiers: all policies, paired bootstrap intervals over matched instances, conditional on three frozen model seeds. Frequency-balanced bins are synthetic unions, not ontological equivalence classes.

Graph	Policy	Native	Coarse	Difference [95% CI]
MANC	Weight	0.0972	0.8226	+0.7254 [+0.7130,+0.7380]
MANC	Random	0.1093	0.9036	+0.7943 [+0.7873,+0.8015]
MANC	Recurrent	0.1121	0.8503	+0.7382 [+0.7294,+0.7474]
MANC	Attribute	0.1088	0.8644	+0.7556 [+0.7446,+0.7667]
MANC	Expert selector	0.1103	0.8629	+0.7527 [+0.7415,+0.7637]
MANC	Random→Attribute	0.1123	0.8741	+0.7618 [+0.7533,+0.7698]
HemiBrain	Weight	0.1290	0.7858	+0.6568 [+0.6438,+0.6702]
HemiBrain	Random	0.1765	0.8887	+0.7122 [+0.7033,+0.7203]
HemiBrain	Recurrent	0.1534	0.8257	+0.6723 [+0.6609,+0.6837]
HemiBrain	Attribute	0.1373	0.8167	+0.6793 [+0.6673,+0.6911]
HemiBrain	Expert selector	0.1409	0.8081	+0.6671 [+0.6545,+0.6798]
HemiBrain	Random→Attribute	0.1477	0.8176	+0.6699 [+0.6599,+0.6799]

Table 10: Constructed certificate mutations, reported as rejected/attempted. Rows are correlated coverage checks of the stated mutation generator, not independent attack samples or estimates of an attack-success probability.

Family	Mutation	H01	MANC	HemiBrain	Total
Provenance/binding	Missing required receipt	800/800	800/800	800/800	2,400/2,400
	Stale episode binding	800/800	800/800	800/800	2,400/2,400
	Wrong graph binding	800/800	800/800	800/800	2,400/2,400
	Wrong query binding	800/800	800/800	800/800	2,400/2,400
	Wrong relation binding	800/800	800/800	800/800	2,400/2,400
Structural	Inconsistent middle node	200/200	200/200	200/200	600/600
	Wrong answer binding	200/200	200/200	200/200	600/600
	Wrong semantic receipt	200/200	200/200	200/200	600/600
Total	–	–	–	–	13,800/13,800

long layouts and retained 221; 32 exceeded the prospective 180-s enrollment timeout and three fell outside the requested length stratum. Inference resamples layouts while retaining their update conditions and timing seeds.

Table 11: Full-episode mechanism audit. Panels are distinct interventions and must not be compared row-wise across panels.

Panel A: irrelevant evidence withdrawal (381 paired schedule cells)							
Method	Success	Actions	Pred.	Succ.	Positions	Lookups	Unsupported
Full replanning	1.000	1170.8	5903.2	17398.8	0.0	0.0	–
Complete validation	1.000	1170.8	62.5	0.0	64.9	0.0	–
Dependency-indexed	1.000	1170.8	0.0	0.0	0.0	1.0	–
Unchecked reuse†	1.000	1170.8	0.0	0.0	0.0	0.0	–

Panel B: final-support loss (513 paired episodes)		
Method	Unsupported authorization episodes	Rate
Full replanning	0/513	0.000
Complete validation	0/513	0.000
Dependency-indexed	0/513	0.000
Unchecked reuse†	511/513	0.996

Pred./Succ. are predicate evaluations and planner-successor expansions. Positions are retained-suffix positions visited. †Unchecked reuse is a safety ablation that performs no validation after an update.